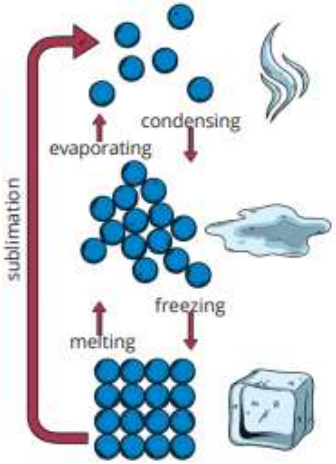
Knowledge Organiser
Year 8: Heating and Cooling – Matter and Thermal Energy

States of matter
- There are three states of matter; solids, liquids and gases
- Solids have particles which are **close together** in a **regular, repeating pattern**
- Liquids have particles which are **close together** but **randomly arranged and moving around.**
- Gases have particles which are **far apart** and moving **quickly around**

Diagram which shows the **changes of state** and the **particle arrangement** of solids, liquids and gases



Changes of state
- When a substance gains or loses enough energy it can change state
- When water reaches 100ºC it will boil, we call this its **boiling point**
- When water reaches 0ºC it will freeze, we call this its **freezing/melting point**

A table which summarises the particle arrangement, volume, shape and ability to flow of solids, liquids and gases

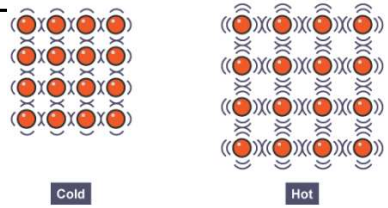
	Solid	Liquid	Gas
particle model diagram			
particle arrangement	regular structure no space between particles	irregular structure very little space between particles	irregular structure large space between particles
volume and shape	fixed volume fixed shape	fixed volume shape changes to fill bottom of container	volume increases to fill capacity shape changes to fill capacity
able to flow	no (forces between particles are very strong and hold them in fixed positions)	yes (forces between particles are weak and particles slide over one another)	yes (forces between particles are very weak and particles move randomly and rapidly)

The table below shows the melting and boiling points of four different gases

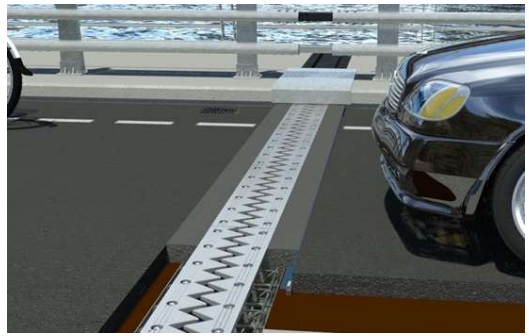
substance	melting point, in °C	boiling point, in °C
carbon dioxide	–78	–78
nitrogen	–210	–196
oxygen	–219	–183
water vapour	0	100

At -185ºC the substances would be in these states of matter:
Carbon dioxide – Solid, it is below the **melting point**
Nitrogen – Gas, it is above the **boiling point**
Oxygen – Liquid, it is above the **melting point** but below the **boiling point**
Water vapour – Gas, it is above the **boiling point**

What happens when particles lose or gain energy?
- When particles gain energy the particles will move more quickly and spread further apart
- When particles lose energy the particles will move more slowly and get closer together
- We say that a substance will **expand** when it gets warmer and **condense** when it gets cooler



This picture shows expansion joints on a bridge which help to stop it from breaking on hot days



How does an expansion joint work?
- On warmer days the metal expands and fills the joint
- If it wasn't there the bridge materials might push against each other and crack

How is temperature different to thermal energy?
- Thermal energy is a measure of the total energy of the particles
- Temperature is a measure of the average kinetic energy of the particles in a substance
- A high temperature makes a substance hotter because the particles gain kinetic energy

RAG Yourself!

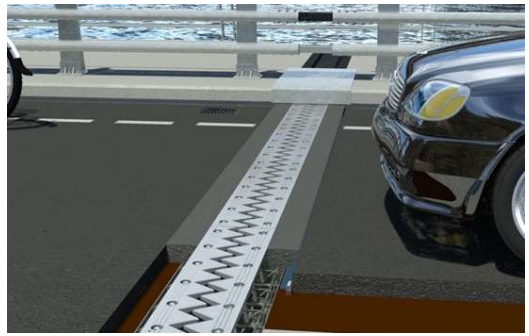
Learning Outcomes;	RAG
Recall the three states of matter and four changes of state	
State the boiling and freezing point of water	
Describe the movement and arrangement of particles in each state	
Explain expansion and contraction	
Describe what happens to particles in a substance when it is heated	
Compare temperature to thermal energy in terms of kinetic energy of particles	



Knowledge Organiser
Year 8: Heating and Cooling – Matter and Thermal Energy

A table which summarises the particle arrangement, volume, shape and ability to flow of solids, liquids and gases

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States of matter

Diagram which shows the **changes of state** and the **particle arrangement** of solids, liquids and gases

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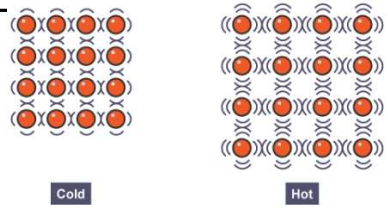
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Changes of state

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